

Hugh Kassir

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EXPERIENCE

Queen Mary, University of London Genomic Data Scientist

London, United Kingdom
Apr 2025 – Aug 2025 (Fixed Term)

- Mapped mouse model phenotypes (IMPC Gene Essentiality) to human rare disorder traits to uncover novel gene-phenotype associations.
- Integrated paralogy and gene essentiality data across >8000 mouse and human genes to investigate genetic redundancy and compensation mechanisms influencing variable disease penetrance and side effects profiles.
- Performed systematic data validation and verification checks to identify and resolve inconsistencies.
- Liaised with IMPC and EBI PIs and partners to investigate discrepancies and ensure accurate interpretation of data for reporting.
- Engineered reproducible data pipelines with Git version control, supporting collaboration with bioinformaticians and researchers.

Postgraduate Research Assistant

Jun 2024 – Dec 2024

- Analysed Genomics England rare disease cohort (>400 patients) to assess genotype-phenotype correlations in neurodevelopmental disorders.
- Conducted statistical analyses (i.e. Fisher's Exact test) to assess the genotype-phenotype relationships, which revealed a statistically significant correlation between the SCN2A gene and delayed social development.
- Integrated Proteomics data to investigate links between variant location in the protein domain and phenotypic severity.
- Developed logistic regression models in R to predict clinical phenotypes from multi-dimensional genomic predictors including gene and variant type.
- Evaluated variants of unknown significance (VUS) across major clinical and population databases (ClinVar, gnomAD, OMIM, and DECIPHER) to determine variant effect and support variant re-classification.

Ezeldin Pharmacy Chief Pharmacist

Beirut, Lebanon
Sep 2019 – Dec 2023

- Supervised two pharmacy staff to ensure accurate dispensing of medications, maintaining compliance with clinical standards and ensuring patient safety.
- Oversaw drug procurement, expenditure tracking, and sales performance analysis to support business development decision and strategic decision-making.
- Produced financial and inventory reports to guide strategic decisions, including optimising stock levels, implementing generic substitutions, and renegotiating supplier contracts.
- Produced and reviewed administrative reports, stock records, and regulatory documentation in line with health data requirements.
- Translated complex therapeutic knowledge into clear recommendations, improving patient satisfaction by 12.3% and adherence.

EDUCATION

MSc Genomic Medicine, Queen Mary University of London (Distinction)

Jan 2024 – Jan 2025

Modules covered include Bioinformatics, Genomics of Common and Rare Diseases, Molecular Pathology of Cancer, Ethical Legal and Social Issues in Genomic Medicine

BS Pharmacy, Lebanese International University

Oct 2011 – Jul 2018

Modules covered include Pharmacology, Pharmacogenomics, Therapeutics.

SKILLS

Statistical Genetics & Bioinformatics: Variant annotation, VUS interpretation, association testing, Variant analysis.

Programming & Data Science: MS Office (Excel, Word, Access), Python, R, PLINK, SQL, Bash, Git, HPC workflows.

Organisational Skills: Meeting support, records management, inventory systems, reproducible data pipelines.

Databases: ClinVar, gnomAD, OMIM, DECIPHER.